1. $$\gcd(a,b)=1, \qquad a \mid bc \quad \Longrightarrow \quad a \mid c$$

   Prove it from Bézout's identity rather than prime factorization.

2. Prove that there are infinitely many primes

   $$p \equiv 3 \mod 4$$

   A Euclid-style argument works, but simply taking the product of known $3 \bmod 4$ primes and adding 1 does not give the useful construction directly. Try to determine what number to construct and why at least one of its prime divisors must be $3 \bmod 4$.

3. Multiplicative structure. Define

   $$d(n) = \#\{d \in \mathbb{N} : d \mid n\}.$$

   Prove that if

   $$\gcd(m,n)=1,$$

   then

   $$d(mn) = d(m)d(n).$$

   Try to formulate this proof as bijection rather than using prime-factorization formula for $d(n)$.